University Physics Test paper of CSU

(2010~2010)

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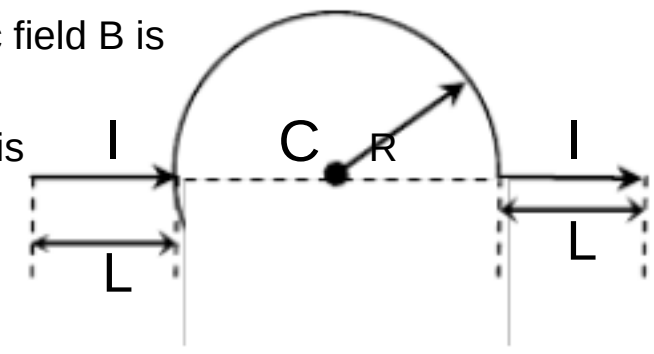
One: Select the right answer in the brackets(4¢ in each problem):

title	1	2	3	4	5	6	7	8	9	10
answers										

1. A solid non-conducting sphere of radius R has a non-uniform charge distribution of volume density $r = r_s r / R$, where r_s is a constant and r is the distance from the center of the sphere. So the total charge on the sphere is

(A) $\frac{4}{3}\pi r_s R^3$; (B) $\pi r_s R^3$ (C) $\pi r_s R^2$; (D) $4\pi r_s R^3$.

2. The wire shown in figure carries current I. So the magnetic field B is produced at the center C of the semicircle by the total wire is



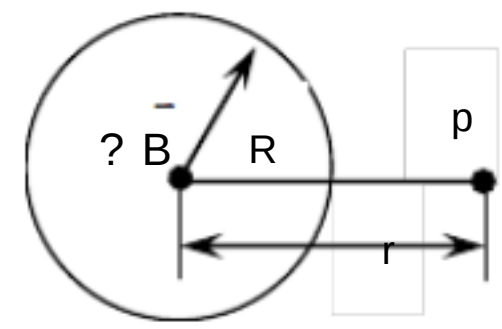
(A) 0; (B) $\mu_0 I / 2R$;

(C) $m_0 I / 4R$; (D) $\mu_0 I / R$.

3. Two concentric thin metal shells, radii are R_1 and R_2 respectively. If the two shells charged q_1 and q_2 and their potentials are U_1 and U_2 respectively. Now if we connect the two shells with conducting wire, their potential is:

(A) U_1 (B) U_2 (C) $U_1 + U_2$ (D) $(U_1 + U_2)/2$

4. As shown in figure ,uniform magnetic field is confined in long cylindrical space .If $\frac{dB}{dt} > 0$,so the magnitude of the induced electric field of point p outside the cylinder E is:



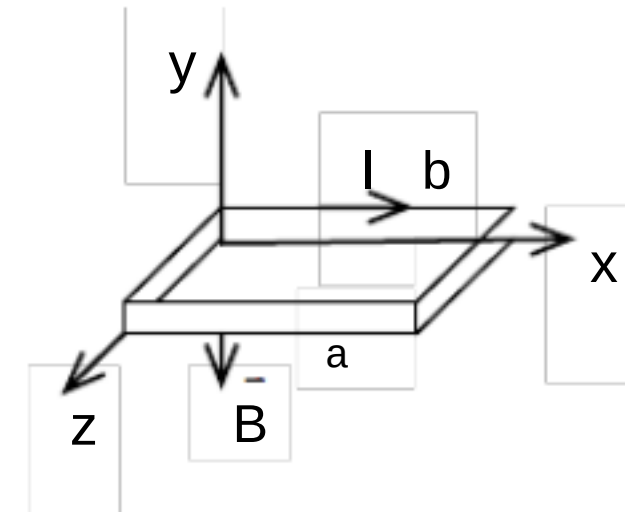
(A) $\frac{R^2}{2r} \times \frac{dB}{dt}$ (B) 0 (C) $\frac{r}{2} \times \frac{dB}{dt}$

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5. The Maxwell equation which describes the variable electric field will produce magnetic field is

(A) $\oint_S D \cdot dS = \int_V \rho \, dV$ (B) $\oint_L E \cdot dl = -\int_S \frac{\partial B}{\partial t} dS$ (C) $\oint_S B \cdot dS = 0$ (D) $\oint_L H \cdot dl = \int_S \left(J + \frac{\partial D}{\partial t} \right) dS$

6. As shown in right figure , the magnetic field is placed in the direction of negative direction of y-axis ,if we know that the electric potential difference between a and b $U_{ab} > 0$, so the material is



(A) N type (B) P type

(C) metal (D) not certain.

7. Which of the following descriptions of wave function is right?

(A) $\Psi(r,t)$ is the probability of finding the particle at position r;

(B) $\Psi(r,t)$ is probability density of finding the particle at position r at time t;

(C) $|\Psi(r,t)|^2$ is probability density of finding the particles at position r at time t;

(D) $|\Psi(r,t)|^2$ is probability of finding the particles at position r at time t.

8. If the quantum number n=1, which of the following quantum state is possible :

(A) (1,1,0,1/2) (B) (1,1,0,0) (C) (1,0,0,-1/2) (D) (1,0,0,0)

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reader	

Two (4 ¢ in each problem): Write the right answer in blanks

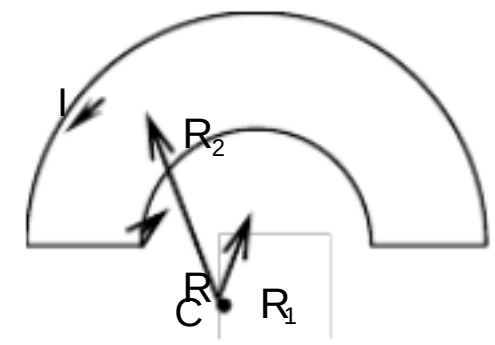
1. The maximum kinetic energy of an electron ejected from a sodium surface whose work function is 2.28 eV when illuminated by light of wavelength 410 nm is _____ eV

2. A circular coil of wire has a diameter of 5.0 cm and contains 25 loops, so the magnetic moment of the coil when it carries a current of 1.5 A is _____ $A \cdot m^2$.

3. If an electron that has velocity $\vec{v} = (2.0 \times 10^6)\vec{i} \quad (3.0 \quad 10^6)\vec{j}$ moves through the magnetic field $\vec{B} = (0.03)\vec{i} - (0.15)\vec{j}$, so the force on the electron is _____.

4. According to Born 's statistical interpretation to wave function, the standard conditions of wave function are: (1)_____ ;(2) _____ _____;(3) _____ _____.

5. A wire is formed into the shape of two half circles connected by equal-length straight sections as shown in figure .A current i flows in the circuit anticlockwise as shown ,the magnetic field at the center C is _____.



6. Two long parallel wires, both of radius a and whose centers are a distance d apart, carry current in opposite directions. Show that ,neglecting the flux(磁通) within the wires ,the inductance (电感)of unit length of a pair of wires is _____.

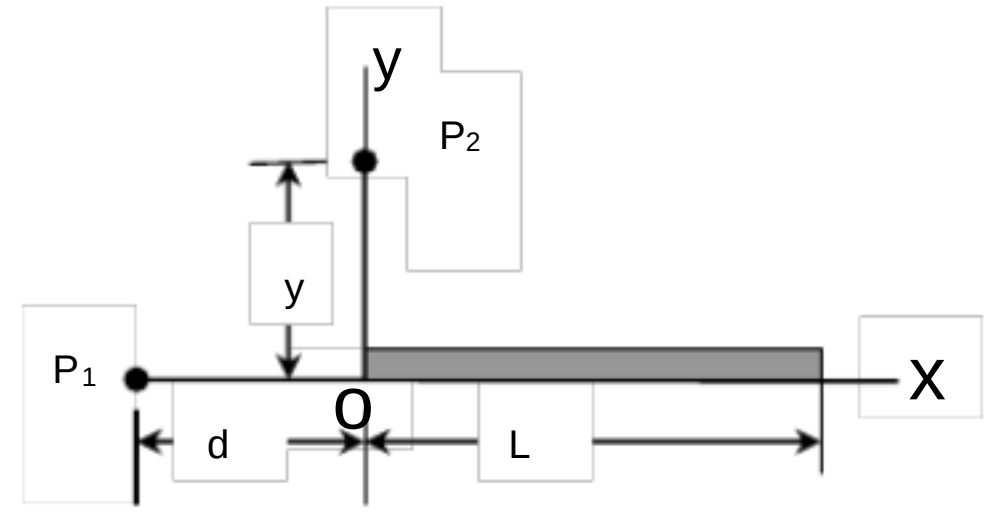
7. The ground-state energy of electron is -13.6eV ,so the energy of the second excited states is _____.

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Three（9分）: The plastic rod shown in figure has length L and a non-uniform linear charge density $l = cx$,where c is a positive constant .With V=0 at infinity, find the electric Potential

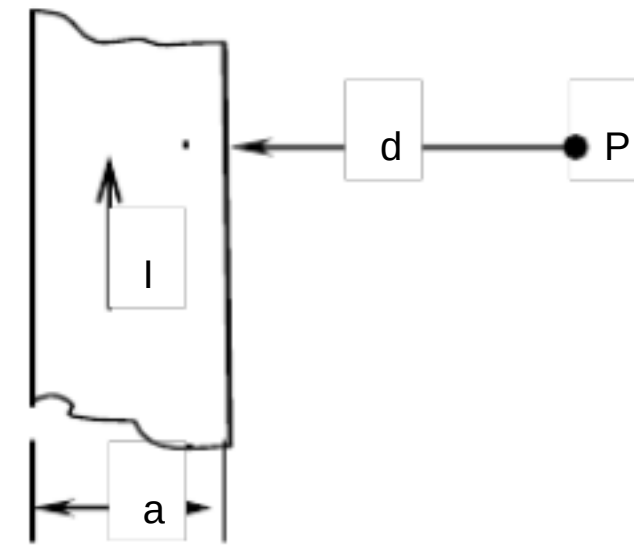
(a)at point p_1 on the x axis, a distance d from one end and

(b) point p_2 on the y axis ,a distance y from one end of the rod.



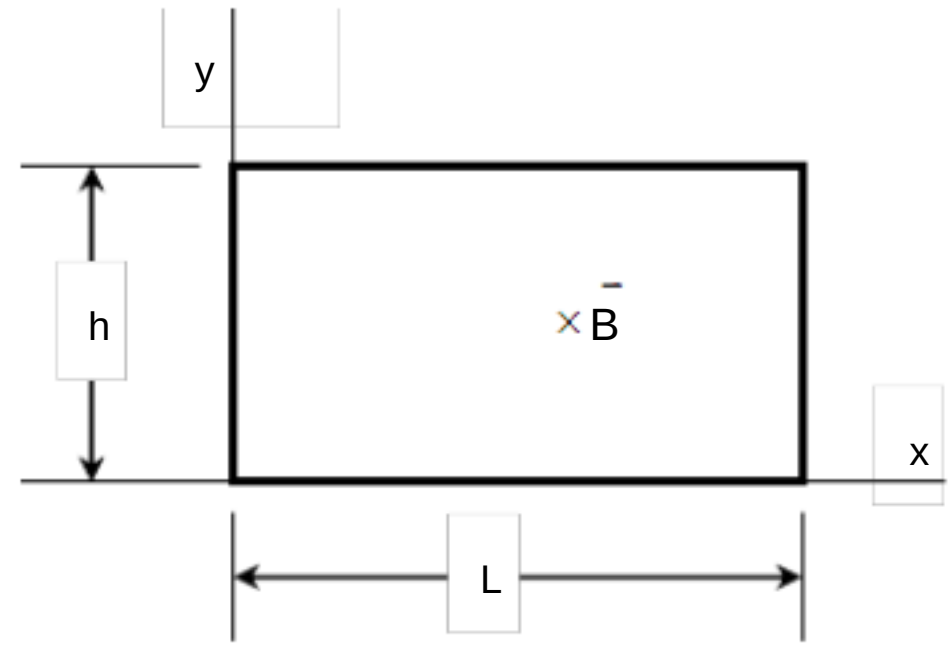
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Four（9分）: As shown in right figure ,an infinite length thin metal plate carries a current I which uniformly distributes in the plate , please work out the magnetic field produced by the current at point P.



Score	
reader	

Five (11分): Figure shows a rectangular loop of wire immersed in a non-uniform and varying magnetic field that is perpendicular to and directed into the page. The field magnitude is given by $B = 400t^2x^2$, with B in Teslas, t in seconds and x in meters. The loop has width $L = 0.3m$ and height $h = 0.2m$. What are the magnitude and direction of the induced emf(感生电动势) around the loop at $t = 10s$?



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Six (11分): The wave function of particles in one dimensional potential well with width a is $y(x) = A\sin(\frac{np}{a}x)$ $(0 \# x \quad a)$,

please work out the following problems:

(1) Normalization constant A;

(2) Where is the maximum probability of finding the particles when n=3?

Appendix: answers for 0813-0815

One: Select the right answer in the brackets (4 ¢ in each problem)

1 B 2 C 3 B 4 A 5 D 6 A 7 C 8 C

Two (4′ **in each blank):** Write the right answer in blanks

1 $0.75eV$ 2 $0.36A \times m^2$ 3 $(6.24 \times 10^{-14} N)\ \vec{k}$ 4 (a) single-value; (b) continuous;

(c) finite 5 $\frac{m_0 I}{4}(\frac{1}{R_1} - \frac{1}{R_2})$ 3 ii 1 6 $L = \frac{m_0}{p} \ln \frac{d-a}{a}$ 7 $-1.51eV$

Three (9 ¢):

Solution: We select point o as the origin and select the element charge $dq = l\,dx$ at the distance x to the origin, so according to the definition of electric potential 2 ¢

(a): The electric potential at point p_1 is

$$U_1 = \int_0^L \frac{dq}{4pe_0(x+d)} = \int_0^L \frac{cxdx}{4pe_0(x+d)} = \frac{c}{4pe_0}(L - d\ln\frac{d+L}{d}) \quad 3¢$$

(b): The electric potential at point p_2 is

$$U_2 = \int_0^L \frac{dq}{4pe_0 r} = \int_0^L \frac{cxdx}{4pe_0\sqrt{x^2+y^2}} = \frac{c}{4pe_0}(\sqrt{L^2+y^2} - y) \quad 4¢$$

Four (9 ¢):

Solution: We select the left side of the plate as the origin and select the element current $di = \frac{I}{a}dx$ (2 ¢) at the distance x to the origin ,according the magnetic field of the infinite long straight wire carrying current:

$$dB = \frac{m_0 di}{2p(a+d-x)} \quad 3¢$$

So: $B = \frac{m_0 I}{2p a}\int_0^a \frac{dx}{a+d-x} = \frac{m_0 I}{2p a}\ln\frac{a+d}{d}$ 2 ¢

Direction: perpendicular the paper inward. 2 ¢

Five (11分):

Solution: Because the magnitude of the magnetic field $\bar{B}$ is changing with the time , the magnetic flux f_B through the loop is also changing .so we can take the differential area $dA = hdx$ (2分)with its normal being parallel to $\bar{B}$, and the flux through the loop is

$$f_B = \int \bar{B} \cdot d\bar{A} = \int B dA = \int Bh dx = 400t^2 x^2 dx \quad 2分$$

take this integration and inserting the integration limits $x = 0$ and $x = 0.3m$ get:

$$f_B = 400t^2 h \int_0^{0.3} x^2 dx = 0.72t^2 \quad 2分, \text{ when } h = 0.2m \text{ ,according to}$$

Farady law at any time :

$$e = \frac{df_B}{dt} = 1.4t^4 \quad 2分$$

For $t = 10s, e = 14.4V$ 2分

Direction: counterclockwise around the loop at this time. 1分

Six (11分)

Solution: **(1)** according to the normalization condition:

$$\int_0^a y^2(x)dx = \int_0^a A^2 \sin^2 \frac{np}{a} x dx = A^2 \sqrt{\frac{a}{2}} = 1 \quad 2分$$

So: $A = \sqrt{\frac{2}{a}}$ 2分

(2) The probability density of finding the particles is:

$y^2(x) = \frac{2}{a} \sin^2 \frac{np}{a} x$, the position of finding the maximum probability satisfies:

$$\frac{dy^2(x)}{dt} = 0 \text{ and } \frac{d^2y^2(x)}{dt^2} < 0 ,$$

We get: $\sin \frac{2np}{a} x = 0$ 2分

Therefore: $\frac{2 \text{ ' } 3p}{a} x = kp \quad (k = 0,1,2\cdots)$

$x = \frac{a}{6} k$, 2分

We can get the position of maximum probability of finding the particles is:

$$x = \frac{a}{6}, \frac{a}{2}, \frac{5a}{6} \quad 3分$$